

## CHAPTER 18—FORECASTING

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### MULTIPLE CHOICE

1. Which of the following is not present in a time series?
  - a. seasonality
  - b. operational variations
  - c. trend
  - d. cycles
2. Given an actual demand of 61, forecast of 58, and an  $\alpha$  of .3, what would the forecast for the next period be using simple exponential smoothing?
  - a. 57.1
  - b. 58.9
  - c. 61.0
  - d. 65.5
3. Which of the following smoothing constants would make an exponential smoothing forecast equivalent to a naive forecast?
  - a. 0
  - b. 1 divided by the number of periods
  - c. 0.5
  - d. 1.0
4. The time series component which reflects a regular, multi-year pattern of being above and below the trend line is
  - a. a trend
  - b. seasonal
  - c. cyclical
  - d. irregular
5. The time series component that reflects variability during a single year is called
  - a. a trend
  - b. seasonal
  - c. cyclical
  - d. irregular
6. The time series component that reflects variability due to natural disasters is called
  - a. a trend
  - b. seasonal
  - c. cyclical
  - d. irregular
7. The time series component that reflects gradual variability over a long time period is called
  - a. a trend
  - b. seasonal
  - c. cyclical
  - d. irregular
8. The trend component is easy to identify by using
  - a. moving averages
  - b. exponential smoothing
  - c. regression analysis
  - d. the Delphi approach

9. The forecasting method that is appropriate when the time series has no significant trend, cyclical, or seasonal effect is
- moving averages
  - mean squared error
  - mean average deviation
  - qualitative forecasting methods
10. If data for a time series analysis is collected on an annual basis only, which component may be ignored?
- trend
  - seasonal
  - cyclical
  - irregular

11. For the following time series, you are given the moving average forecast.

| Time Period | Time Series Value | Moving Average Forecast |
|-------------|-------------------|-------------------------|
| 1           | 23                |                         |
| 2           | 17                |                         |
| 3           | 17                |                         |
| 4           | 26                | 19                      |
| 5           | 11                | 20                      |
| 6           | 23                | 18                      |
| 7           | 17                | 20                      |

The mean squared error equals

- 0
  - 6
  - 41
  - 164
12. If the estimate of the trend component is 158.2, the estimate of the seasonal component is 94%, the estimate of the cyclical component is 105%, and the estimate of the irregular component is 98%, then the multiplicative model will produce a forecast of
- 1.53
  - 1.53%
  - 153.02
  - 153,020,532
13. Below you are given the first four values of a time series.

| Time Period | Time Series Value |
|-------------|-------------------|
| 1           | 18                |
| 2           | 20                |
| 3           | 25                |
| 4           | 17                |

Using a 4-period moving average, the forecasted value for period 5 is

- 2.5
  - 17
  - 20
  - 10
14. Below you are given the first two values of a time series. You are also given the first two values of the exponential smoothing forecast.

| Time Period (t) | Time Series Value (Y <sub>t</sub> ) | Exponential Smoothing Forecast (F <sub>t</sub> ) |
|-----------------|-------------------------------------|--------------------------------------------------|
| 1               | 18                                  | 18                                               |
| 2               | 22                                  | 18                                               |

If the smoothing constant equals .3, then the exponential smoothing forecast for time period three is

- a. 18
- b. 19.2
- c. 20
- d. 40

15. The following linear trend expression was estimated using a time series with 17 time periods.

$$T_t = 129.2 + 3.8t$$

The trend projection for time period 18 is

- a. 68.4
- b. 193.8
- c. 197.6
- d. 6.84

NARRBEGIN: Exhibit 18-1

Exhibit 18-1

Below you are given the first five values of a quarterly time series. The multiplicative model is appropriate and a four-quarter moving average will be used.

| Year | Quarter | Time Series Value $Y_t$ |
|------|---------|-------------------------|
| 1    | 1       | 36                      |
|      | 2       | 24                      |
|      | 3       | 16                      |
| 2    | 4       | 20                      |
|      | 1       | 44                      |

NARREND

16. Refer to Exhibit 18-1. An estimate of the trend component times the cyclical component ( $T_2C_t$ ) for Quarter 3 of Year 1, when a four-quarter moving average is used, is
- a. 24
  - b. 25
  - c. 26
  - d. 28
17. Refer to Exhibit 18-1. An estimate of the seasonal-irregular component for Quarter 3 of Year 1 is
- a. .64
  - b. 1.5625
  - c. 5.333
  - d. 30
18. You are given the following information on the seasonal-irregular component values for a quarterly time series:

| Quarter | Seasonal-Irregular<br>Component Values ( $S_tI_t$ ) |
|---------|-----------------------------------------------------|
| 1       | 1.23, 1.15, 1.16                                    |
| 2       | .86, .89, .83                                       |
| 3       | .77, .72, .79                                       |
| 4       | 1.20, 1.13, 1.17                                    |

The seasonal index for Quarter 1 is

- a. .997
- b. 1.18
- c. 4
- d. 3

19. Below you are given some values of a time series consisting of 26 time periods.

| Time Period | Time Series Value |
|-------------|-------------------|
| 1           | 37                |
| 2           | 48                |
| 3           | 50                |
| 4           | 63                |
| .           |                   |
| .           |                   |
| .           |                   |
| 23          | 105               |
| 24          | 107               |
| 25          | 112               |
| 26          | 114               |

The estimated regression equation for these data is

$$Y_t = 16.23 + .52Y_{t-1} + .37Y_{t-2}$$

The forecasted value for time period 27 is

- 53.23
  - 109.5
  - 116.65
  - 116.95
20. A group of observations measured at successive time intervals is known as
- a trend component
  - a time series
  - a forecast
  - an additive time series model
21. A component of the time series model that results in the multi-period above-trend and below-trend behavior of a time series is
- a trend component
  - a cyclical component
  - a seasonal component
  - an irregular component
22. The model that assumes that the actual time series value is the product of its components is the
- forecast time series model
  - multiplicative time series model
  - additive time series model
  - None of these alternatives is correct.
23. A method of smoothing a time series that can be used to identify the combined trend/cyclical component is
- the moving average
  - the percent of trend
  - exponential smoothing
  - the trend/cyclical index
24. A method that uses a weighted average of past values for arriving at smoothed time series values is known as
- a smoothing average
  - a moving average
  - an exponential average
  - an exponential smoothing

25. In the linear trend equation,  $T = b_0 + b_1t$ ,  $b_1$  represents the
  - a. trend value in period  $t$
  - b. intercept of the trend line
  - c. slope of the trend line
  - d. point in time
26. In the linear trend equation,  $T = b_0 + b_1t$ ,  $b_0$  represents the
  - a. time
  - b. slope of the trend line
  - c. trend value in period 1
  - d. the Y intercept
27. A parameter of the exponential smoothing model which provides the weight given to the most recent time series value in the calculation of the forecast value is known as the
  - a. mean square error
  - b. mean absolute deviation
  - c. smoothing constant
  - d. None of these alternatives is correct.
28. One measure of the accuracy of a forecasting model is
  - a. the smoothing constant
  - b. a deseasonalized time series
  - c. the mean square error
  - d. None of these alternatives is correct.
29. A qualitative forecasting method that obtains forecasts through "group consensus" is known as the
  - a. Autoregressive model
  - b. Delphi approach
  - c. mean absolute deviation
  - d. None of these alternatives is correct.

NARRBEGIN: Exhibit 18-2

Exhibit 18-2

Consider the following time series.

|       |   |   |   |    |
|-------|---|---|---|----|
| $t$   | 1 | 2 | 3 | 4  |
| $Y_t$ | 4 | 7 | 9 | 10 |

NARREND

30. Refer to Exhibit 18-2. The slope of linear trend equation,  $b_1$ , is
  - a. 2.5
  - b. 2.0
  - c. 1.0
  - d. 1.25
31. Refer to Exhibit 18-2. The intercept,  $b_0$ , is
  - a. 2.5
  - b. 2.0
  - c. 1.0
  - d. 1.25

32. Refer to Exhibit 18-2. The forecast for period 5 is
- 10.0
  - 2.5
  - 12.5
  - 4.5
33. Refer to Exhibit 18-2. The forecast for period 10 is
- 10.0
  - 25.0
  - 30.0
  - 22.5

NARRBEGIN: Exhibit 18-3

Exhibit 18-3

Consider the following time series.

| Year (t) | $Y_t$ |
|----------|-------|
| 1        | 7     |
| 2        | 5     |
| 3        | 4     |
| 4        | 2     |
| 5        | 1     |

NARREND

34. Refer to Exhibit 18-3. The slope of linear trend equation,  $b_1$ , is
- 1.5
  - +1.5
  - 8.3
  - 8.3
35. Refer to Exhibit 18-3. The intercept,  $b_0$ , is
- 1.5
  - +1.5
  - 8.3
  - 8.3
36. Refer to Exhibit 18-3. In which time period does the value of  $Y_t$  reach zero?
- 0.000
  - 0.181
  - 5.53
  - 4.21
37. Refer to Exhibit 18-3. The forecast for period 10 is
- 6.7
  - 6.7
  - 23.3
  - 15

## PROBLEM

- Consider the following time series.

| Time | $Y_t$ |
|------|-------|
| 1    | 18    |
| 2    | 20    |
| 3    | 22    |
| 4    | 24    |
| 5    | 26    |
| 6    | 28    |

- Develop a linear trend equation for this time series.
  - What is the forecast for  $t = 17$ ?
- What is the forecast for July based on a three-month weighted moving average applied to the following past demand data and using the weights: 5, 3, and 2 (largest weight is for most recent data)?  
*Show all of your computations for April through July.*

| Month    | Demand | Forecast |
|----------|--------|----------|
| January  | 40     |          |
| February | 45     |          |
| March    | 57     |          |
| April    | 60     |          |
| May      | 75     |          |
| June     | 87     |          |
| July     |        |          |

- What is the forecast for July based on a three-month weighted moving average applied to the following past demand data and using the weights: 6, 4, and 2 (largest weight is for most recent data)?  
*Show all of your computations for April through July.*

| Month    | Actual Demand (A) |
|----------|-------------------|
| January  | 40                |
| February | 45                |
| March    | 57                |
| April    | 60                |
| May      | 75                |
| June     | 87                |
| July     |                   |

- What is the forecast for July based on a three-month weighted moving average applied to the following past demand data and using the weights: 5, 4, and 3 (largest weight is for most recent data)?  
*Show all of your computations for April through July.*

| Month    | Demand |
|----------|--------|
| January  | 80     |
| February | 83     |
| March    | 87     |
| April    | 90     |
| May      | 95     |
| June     | 98     |
| July     |        |

5. Actual sales for January through April are shown below.

| Observation | Month    | Actual Sales<br>(A) | Forecasted Sales<br>(F) |
|-------------|----------|---------------------|-------------------------|
| 1           | January  | 18                  |                         |
| 2           | February | 23                  |                         |
| 3           | March    | 20                  |                         |
| 4           | April    | 16                  |                         |
| 5           | May      |                     |                         |

Use exponential smoothing with  $\alpha = 0.2$  to calculate smoothed averages and forecast sales for May from the above data. Assume the forecast for the initial period (January) is 18. *Show all of your computations.*

6. Actual sales for January through April are shown below.

| Month    | Actual Sales (A) |
|----------|------------------|
| January  | 18               |
| February | 25               |
| March    | 30               |
| April    | 40               |
| May      |                  |

Use exponential smoothing with  $\alpha = 0.3$  to calculate smoothed values and forecast sales for May from the above data. Assume the forecast for the initial period (January) is 18. *Show all of your computations from February through May.*

7. Actual sales of a company (in millions of dollars) for January through April are shown below.

| Month    | Sales |
|----------|-------|
| January  | 18    |
| February | 25    |
| March    | 30    |
| April    | 40    |
| May      |       |

- Use  $\alpha = 0.3$  to compute the exponential smoothing values for sales. Compute MSE and forecast sales for May. *Show all of your computations from February through May.*
  - Use  $\alpha = 0.1$  to compute the exponential smoothing values for sales. Compute MSE and forecast sales for May. *Show all of your computations from February through May.*
  - Based on MSE, which  $\alpha$  provides a better forecast? Explain why?
8. The actual demand for a product and the forecast for the product are shown below. Calculate MAE and MSE. *Show all of your computations.*

| Observation | Actual Demand<br>(A) | Forecast<br>(F) |
|-------------|----------------------|-----------------|
| 1           | 35                   | ---             |
| 2           | 30                   | 35              |
| 3           | 26                   | 30              |
| 4           | 34                   | 26              |
| 5           | 28                   | 34              |
| 6           | 38                   | 28              |

9. Demand for a product and the forecasting department's forecast (naïve model) for a product are shown below.

- Compute the mean absolute error.
- Compute the mean squared error.

| Period | Actual Demand (A) | Forecasted Demand (F) |
|--------|-------------------|-----------------------|
| 1      | 45                | ---                   |
| 2      | 48                | 45                    |
| 3      | 42                | 48                    |
| 4      | 44                | 42                    |
| 5      | 50                | 44                    |
| 6      | 60                | 50                    |

10. For the following time series data, using the naïve method (the most recent value as the forecast for the next period), compute the following measures of forecast accuracy.

| Month | Value |
|-------|-------|
| 1     | 12    |
| 2     | 14    |
| 3     | 10    |
| 4     | 16    |
| 5     | 29    |
| 6     | 22    |

- Mean absolute error (MAE)
- Mean squared error (MSE)
- What is the forecast for period 7?

11. The quarterly sales of a company (in millions of dollars) over the past three years are given in the following table.

|           | 2007 | 2008 | 2009 |
|-----------|------|------|------|
| Quarter 1 | 170  | 180  | 190  |
| Quarter 2 | 111  | 96   | 120  |
| Quarter 3 | 270  | 280  | 290  |
| Quarter 4 | 250  | 220  | 223  |

- Compute the four seasonal factors (Seasonal Indexes). *Show all of your computations.*
- The trend for these data is  $\text{Trend} = 174 + 4t$  ( $t$  represents time, where  $t=1$  for Quarter 1 of 2007 and  $t=12$  for Quarter 4 of 2009). Forecast sales for the first quarter of 2010 using the *trend and seasonal indexes*. *Show all of your computations.*

12. The quarterly sales of a company (*in millions of dollars*) over the past three years are given in the following table.

|           | 2007 | 2008 | 2009 |
|-----------|------|------|------|
| Quarter 1 | 106  | 135  | 149  |
| Quarter 2 | 256  | 280  | 292  |
| Quarter 3 | 273  | 280  | 290  |
| Quarter 4 | 190  | 180  | 209  |

- Compute the four seasonal factors (Seasonal Indexes). *Show all of your computations.*
  - The trend for these data is  $\text{Trend} = 185.86 + 5.25 t$  ( $t$  represents time, where  $t=1$  for Quarter 1 of 2007 and  $t=12$  for Quarter 4 of 2009). Forecast sales for the first quarter of 2010 using the *trend only*. *Show all of your computations.*
  - Forecast sales for the first quarter of 2010 using the *trend and seasonal indexes* and write your answer below. *Show all of your computations.*
13. The quarterly sales of a company (*in millions of dollars*) over the past three years are given in the following table.

| Quarter   | 2007 | 2008 | 2009 |
|-----------|------|------|------|
| Quarter 1 | 14   | 28   | 30   |
| Quarter 2 | 20   | 16   | 18   |
| Quarter 3 | 36   | 40   | 38   |
| Quarter 4 | 10   | 14   | 12   |

- Compute the four seasonal factors (Seasonal Indexes). *Show all of your computations.*
  - The trend for these data is  $\text{Trend} = 20.82 + 0.336 t$  ( $t$  represents time, where  $t=1$  for Quarter 1 of 2007 and  $t=12$  for Quarter 4 of 2009). Forecast sales for the first quarter of 2010 using the *trend only*. *Show all of your computations.*
  - Forecast sales for the first quarter of 2010 using the trend and seasonal indexes and write your answer below. *Show all of your computations.*
14. The sales records of a company over a period of seven years are shown below.

| Year<br>( $t$ ) | Sales<br>(In Millions of Dollars) |
|-----------------|-----------------------------------|
| 1               | 12                                |
| 2               | 16                                |
| 3               | 17                                |
| 4               | 19                                |
| 5               | 18                                |
| 6               | 21                                |
| 7               | 22                                |

- Develop a linear trend expression for the above time series.
- Forecast sales for period 10.

15. Student enrollment at a university over the past six years is given below.

| Year<br>(t) | Enrollment<br>(In 1,000s) |
|-------------|---------------------------|
| 1           | 6.30                      |
| 2           | 7.70                      |
| 3           | 8.00                      |
| 4           | 8.20                      |
| 5           | 8.80                      |
| 6           | 8.00                      |

- Develop a linear trend expression for the above time series.
- Forecast enrollment for year 10.

16. The following time series shows the sales of a clothing store over a 10-week period.

| Week | Sales<br>(\$1,000s) |
|------|---------------------|
| 1    | 15                  |
| 2    | 16                  |
| 3    | 19                  |
| 4    | 18                  |
| 5    | 19                  |
| 6    | 20                  |
| 7    | 19                  |
| 8    | 22                  |
| 9    | 15                  |
| 10   | 21                  |

- Compute a 4-week moving average for the above time series.
- Compute the mean square error (MSE) for the 4-week moving average forecast.
- Use  $\alpha = 0.3$  to compute the exponential smoothing values for the time series.
- Forecast sales for week 11.

17. The following time series shows the number of units of a particular product sold over the past six months.

| Month | Units Sold<br>(Thousands) |
|-------|---------------------------|
| 1     | 8                         |
| 2     | 3                         |
| 3     | 4                         |
| 4     | 5                         |
| 5     | 12                        |
| 6     | 10                        |

- Compute a 3-month moving average (centered) for the above time series.
- Compute the mean square error (MSE) for the 3-month moving average.
- Use  $\alpha = 0.2$  to compute the exponential smoothing values for the time series.
- Forecast the sales volume for month 7.

18. The sales volumes of CMM, Inc., a computer firm, for the past 8 years is given below.

| Year<br>(t) | Sales<br>(In Millions of Dollars) |
|-------------|-----------------------------------|
| 1           | 2                                 |
| 2           | 3                                 |
| 3           | 5                                 |
| 4           | 4                                 |
| 5           | 6                                 |
| 6           | 8                                 |
| 7           | 9                                 |
| 8           | 9                                 |

- a. Develop a linear trend expression for the above time series.
- b. Forecast sales for period 9.

19. The sales records of a major auto manufacturer over the past ten years are shown below.

| Year (t) | Number of Cars Sold<br>(In thousands of Units) |
|----------|------------------------------------------------|
| 1        | 195                                            |
| 2        | 200                                            |
| 3        | 250                                            |
| 4        | 270                                            |
| 5        | 320                                            |
| 6        | 380                                            |
| 7        | 440                                            |
| 8        | 460                                            |
| 9        | 500                                            |
| 10       | 500                                            |

Develop a linear trend expression and project the sales (the number of cars sold) for time period  $t = 11$ .

20. The following data show the quarterly sales of Amazing Graphics, Inc. for the years 6 through 8.

| Year | Quarter | Sales |
|------|---------|-------|
| 6    | 1       | 2.5   |
|      | 2       | 1.5   |
|      | 3       | 2.4   |
|      | 4       | 1.6   |
| 7    | 1       | 2.0   |
|      | 2       | 1.4   |
|      | 3       | 1.7   |
|      | 4       | 1.9   |
| 8    | 1       | 2.5   |
|      | 2       | 2.0   |
|      | 3       | 2.4   |
|      | 4       | 2.1   |

- a. Compute the four-quarter moving average values for the above time series.
- b. Compute the seasonal factors for the four quarters.
- c. Use the seasonal factors developed in Part b to adjust the forecast for the effect of season for year 6.

21. John has collected the following information on the amount of tips he has collected from parking cars the last seven nights.

| Day | Tips |
|-----|------|
| 1   | 18   |
| 2   | 22   |
| 3   | 17   |
| 4   | 18   |
| 5   | 28   |
| 6   | 20   |
| 7   | 12   |

- a. Compute the 3-day moving averages for the time series.
- b. Compute the mean square error for the forecasts.
- c. Compute the mean absolute deviation for the forecasts.
- d. Forecast John's tips for day 7.

22. The following information has been collected on the sales of greeting cards for the past 6 weeks.

| Week | Sales |
|------|-------|
| 1    | 105   |
| 2    | 90    |
| 3    | 95    |
| 4    | 110   |
| 5    | 105   |
| 6    | 100   |

- Produce exponential smoothing forecasts for the series using a smoothing constant of .2.
  - Compute the mean square error for the forecasts produced with a smoothing constant of .2.
  - What is the forecast of sales for week 7?
  - Is a smoothing constant of .2 or .3 better for the sales data? Explain.
23. Consider the following annual series on the number of people assisted by a county human resources department.

| Year | People (in 100s) |
|------|------------------|
| 1    | 22               |
| 2    | 24               |
| 3    | 28               |
| 4    | 24               |
| 5    | 22               |
| 6    | 24               |
| 7    | 20               |
| 8    | 26               |
| 9    | 24               |
| 10   | 28               |
| 11   | 26               |

- Prepare 3-year moving average values to be used as forecasts for periods 4 through 11. Calculate the mean squared error (MSE) measure of forecast accuracy for periods 4 through 11.
  - Use a smoothing constant of .4 to compute exponential smoothing values to be used as forecasts for periods 2 through 11. Calculate the MSE.
  - Compare the results in Parts a and b.
24. The temperature in Chicago has been recorded for the past seven days. You are given the information below.

| Day | Temperature |
|-----|-------------|
| 1   | 82          |
| 2   | 80          |
| 3   | 84          |
| 4   | 83          |
| 5   | 80          |
| 6   | 79          |
| 7   | 82          |

- Produce exponential smoothing forecasts for the series using a smoothing constant of .2.
  - Compute the mean square error for the forecasts produced with a smoothing constant of .2.
  - What is the forecasted temperature for day 8?
  - Is a smoothing constant of .2 or .3 better for the temperature data? Explain.
25. The yearly series below exhibits a long-term trend. Use the appropriate forecasting technique to produce forecasts for years 11 and 12.

| Year | Time Series Value |
|------|-------------------|
| 1    | 120               |
| 2    | 132               |
| 3    | 148               |
| 4    | 152               |
| 5    | 160               |
| 6    | 175               |
| 7    | 182               |
| 8    | 190               |
| 9    | 195               |
| 10   | 205               |

26. The following time series gives the number of units sold during 5 years at a boat dealership.

| Year | Quarter | Number of Units |
|------|---------|-----------------|
| 1    | 1       | 300             |
|      | 2       | 240             |
|      | 3       | 240             |
|      | 4       | 290             |
| 2    | 1       | 350             |
|      | 2       | 300             |
|      | 3       | 280             |
|      | 4       | 320             |
| 3    | 1       | 410             |
|      | 2       | 400             |
|      | 3       | 390             |
|      | 4       | 410             |
| 4    | 1       | 490             |
|      | 2       | 450             |
|      | 3       | 440             |
|      | 4       | 510             |
| 5    | 1       | 540             |
|      | 2       | 530             |
|      | 3       | 520             |
|      | 4       | 540             |

- Find the four-quarter centered moving averages.
  - Plot the series and the moving averages on a graph.
  - Compute the seasonal-irregular component.
  - Compute the seasonal factors for all four quarters.
  - Compute the deseasonalized time series for sales.
  - Calculate the linear trend from the deseasonalized sales.
  - Forecast the number of units sold in each quarter of year 6.
27. Below you are given information on John's income for the past 7 years.
- | Year | Income (In Thousands) |
|------|-----------------------|
| 1    | 15.0                  |
| 2    | 16.2                  |
| 3    | 17.1                  |
| 4    | 18.1                  |
| 5    | 18.8                  |
| 6    | 19.2                  |
| 7    | 20.5                  |
- Use regression analysis to obtain an expression for the linear trend component.
  - Forecast John's income for the next 5 years.

28. You are given the following information on the quarterly profits for Ajax Corporation.

| Year | Quarter | Quarterly Profits $Y_t$ |
|------|---------|-------------------------|
| 1    | 1       | 150                     |
|      | 2       | 120                     |
|      | 3       | 160                     |
|      | 4       | 150                     |
| 2    | 1       | 150                     |
|      | 2       | 130                     |
|      | 3       | 180                     |
|      | 4       | 160                     |
| 3    | 1       | 170                     |
|      | 2       | 140                     |
|      | 3       | 200                     |
|      | 4       | 180                     |
| 4    | 1       | 200                     |
|      | 2       | 150                     |
|      | 3       | 230                     |
|      | 4       | 200                     |

- Find the four-quarter centered moving averages.
- Compute the seasonal-irregular component.
- Compute the seasonal factors for all four quarters.
- Represent the deseasonalized series.

29. Below you are given information on crime statistics for Middletown.

| Year | Quarter | Number of Crimes Committed $Y_t$ |
|------|---------|----------------------------------|
| 1    | 1       | 10                               |
|      | 2       | 20                               |
|      | 3       | 25                               |
|      | 4       | 5                                |
| 2    | 1       | 10                               |
|      | 2       | 30                               |
|      | 3       | 35                               |
|      | 4       | 25                               |
| 3    | 1       | 20                               |
|      | 2       | 40                               |
|      | 3       | 35                               |
|      | 4       | 15                               |
| 4    | 1       | 20                               |
|      | 2       | 50                               |
|      | 3       | 45                               |
|      | 4       | 35                               |

The seasonal factors for these data are

| Quarter | Seasonal Factor $S_t$ |
|---------|-----------------------|
| 1       | .589                  |
| 2       | 1.351                 |
| 3       | 1.335                 |
| 4       | .726                  |

- Deseasonalize the series.
- Obtain an estimate of the linear trend for this series.
- Use the seasonal and trend components to forecast the number of crimes for each quarter of Year 5.

30. Below you are given the seasonal factors and the estimated trend equation for a time series. These values were computed on the basis of 5 years of quarterly data.

| Quarter | Seasonal Factor $S_t$ |
|---------|-----------------------|
| 1       | 1.2                   |
| 2       | .9                    |
| 3       | .8                    |
| 4       | 1.1                   |

$$T = 126.23 - 1.6t$$

Produce forecasts for all four quarters of year 6 by using the seasonal and trend components.

31. The following data show the quarterly sales of a major auto manufacturer (introduced in exercise 4) for the years 8 through 10.

| Year | Quarter | Sales |
|------|---------|-------|
| 8    | 1       | 160   |
|      | 2       | 180   |
|      | 3       | 190   |
|      | 4       | 170   |
| 9    | 1       | 200   |
|      | 2       | 210   |
|      | 3       | 260   |
|      | 4       | 230   |
| 10   | 1       | 210   |
|      | 2       | 240   |
|      | 3       | 290   |
|      | 4       | 260   |

- Compute the four-quarter moving average values for the above time series.
- Compute the seasonal factors for the four quarters.
- Use the seasonal factors developed in Part b to adjust the forecast for the effect of season for year 9.